Representation of passband random processes

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Baseband and passband stochastic processes

- We will henceforth assume real wide-sense stationary (WSS) stochastic processes, with zero mean.
- A process $x_b(t)$ is said to be *baseband* if its power spectral density (PSD) $S_{x_b x_b}(f)$ is baseband, namely:
$$S_{x_b x_b}(f) = 0, \quad \text{if } |f| > W > 0$$
- A process $x_p(t)$ is said to be *passband* if its PSD $S_{x_p x_p}(f)$ is passband, namely:
$$S_{x_p x_p}(f) = 0, \quad \text{if } |f \pm f_c| > W > 0$$
- We are interested in determining a suitable representation of passband stochastic processes. Reasoning as in the deterministic case, we construct the random process
$$x_p^+(t) = \frac{1}{\sqrt{2}} x_p(t) + \frac{j}{\sqrt{2}} \widehat{x}_p(t)$$
where $\widehat{x}_p(t)$ is obtained by filtering $x_p(t)$ through the Hilbert filter having impulse response $h(t) = 1/(\pi t)$ or, equivalently, frequency response $H(f) = -j\text{sgn}(f)$.

Rappresentazione dei processi aleatori passa banda

Lavoriamo con processi aleatori:

- Reali + Ipotesi non troppo restrittiva
- Stazionari in senso Lato WSS
- a media nulla ($\mu_X = E[X(t)] = 0$)

Un processo aleatorio è detto in banda base se la sua Densità spettrale di Potenza (PSD) $S_{X_b X_b}(f)$ è limitata intorno all'origine

$$S_{X_b X_b}(f) = 0 \qquad \forall f: |f| > W > 0$$

Un processo aleatorio è detto in banda passante se la sua PSD $S_{X_p X_p}(f)$ è concentrata esclusivamente attorno alle frequenze $\pm f_c$ con $f_c \gg W$

$$S_{X_p X_p}(f) = 0 \qquad \forall f: \; |\,|f| - f_c\,| > W > 0$$

Posso costruire il PROCESSO COMPLESSO ANALITICO, per eliminare la ridondanza delle frequenze negative del processo banda passante

come fatto con i segnali deterministici.

$$X_p^+(t) = \frac{1}{\sqrt{2}} x_p(t) + \frac{j}{\sqrt{2}} \hat{x}_p(t) \quad \leftarrow \text{segnale ottenuto considerando unicamente lo spettro a frequenze positive}$$

dove: . $\hat{X}_p(t)$ è la TRASFORMATA DI HILBERT del processo $X(t)$

ottenuta attraverso un filtro di Hilbert ideale con risposta impulsiva $h(t) = \frac{1}{\pi t}$ e risposta in frequenza:

$$H(f) = -j\,\mathrm{sgn}(f) = \begin{cases} -j & f > 0 \\ j & f < 0 \end{cases}$$

Representation of passband stochastic processes (1/4)

- Parallel to the deterministic case, we define the *complex envelope* $x(t)$ of the process $x_p(t)$ as:

$$x(t) = x_p^+(t)\, e^{-j2\pi f_c t} = x_c(t) + j x_s(t) \tag{1}$$

  where, as in the deterministic case, $x_c(t)$ and $x_s(t)$ are called *in-phase* and *quadrature* components of the process $x_p(t)$, respectively.

- The usual relation between the passband signal $x_p(t)$ and complex envelope $x(t)$ holds:

$$x_p(t) = \sqrt{2}\, x_c(t)\, \cos(2\pi\, f_c\, t) - \sqrt{2}\, x_s(t)\, \sin(2\pi\, f_c\, t)$$

  with

$$x_c(t) = \frac{1}{\sqrt{2}} \left[x_p(t)\, \cos(2\pi\, f_c\, t) + \widehat{x}_p(t)\, \sin(2\pi\, f_c\, t) \right]$$

$$x_s(t) = \frac{1}{\sqrt{2}} \left[-x_p(t)\, \sin(2\pi\, f_c\, t) + \widehat{x}_p(t)\, \cos(2\pi\, f_c\, t) \right]$$

In modo omologo a quanto fatto per i segnali deterministici definiamo L'INVILUPPO COMPLESSO del processo aleatorio

- L'INVILUPPO COMPLESSO del processo aleatorio $X_p(t)$ è definito:

$$X(t) \triangleq X_p^+(t) e^{-j2\pi f_c t} = X_c(t) + jX_s(t)$$

↑ consiste in una traslazione in frequenza del processo che centra la PSD nell'origine

$X_c(t)$: componente in fase di $X_p(t)$

$X_s(t)$: componente in quadratura di $X_p(t)$

Sia $X_c(t)$ sia $X_s(t)$ sono processi aleatori reali in banda base

La relazione tra il processo aleatorio banda passante $X_p(t)$ e l'inviluppo complesso $X(t)$ è:

$$X_p(t) = \sqrt{2}\,\mathrm{Re}\{X_p^+(t)\} = \sqrt{2}\,\mathrm{Re}\{X(t)e^{j2\pi f_c t}\} = \sqrt{2}X_c(t)\cos(2\pi f_c t) - \sqrt{2}X_s(t)\sin(2\pi f_c t)$$

$\left[X_p^+(t) = \frac{1}{\sqrt{2}}X_p(t) + \frac{j}{\sqrt{2}}\hat{X}_p(t)\right]$

questa espressione caratterizza la legame tra $X_p(t)$ e $X(t) = X_c(t) + jX_s(t)$

Estrazione delle componenti $X_c(t)$ e $X_s(t)$:

$$X(t) = X_c(t) + jX_s(t) = X_p^+(t)e^{-j2\pi f_c t} = \left[\frac{1}{\sqrt{2}}X_p(t) + j\cdot\frac{1}{\sqrt{2}}\hat{X}_p(t)\right]e^{-j2\pi f_c t}$$

- $X_c(t) = \mathrm{Re}\{X(t)\} = \mathrm{Re}\{X_p^+(t)e^{-j2\pi f_c t}\} = \frac{1}{\sqrt{2}}\left[X_p(t)\cos(2\pi f_c t) + \hat{X}_p(t)\sin(2\pi f_c t)\right]$
- $X_s(t) = \mathrm{Im}\{X(t)\} = \mathrm{Im}\{X_p^+(t)e^{-j2\pi f_c t}\} = \frac{1}{\sqrt{2}}\left[-X_p(t)\sin(2\pi f_c t) + \hat{X}_p(t)\cos(2\pi f_c t)\right]$

Representation of passband stochastic processes (2/4)

- **Theorem**: If $x_p(t)$ is a zero-mean WSS bandpass process, the processes $x_c(t)$ and $x_s(t)$ will be zero-mean jointly WSS processes.

  Proof: It is plain to prove that both $x_c(t)$ and $x_s(t)$ are zero-mean.
  It remains to verify that the autocorrelation functions $R_{x_c x_c}(t,\tau)$ and $R_{x_s x_s}(t,\tau)$, and the crosscorrelation function $R_{x_c x_s}(t,\tau)$ depend on τ only.
  Remembering the input-output relationships between correlation functions (see lectures on random processes) and invoking the Wiener-Khinchin theorem, we may first derive the following relations:

$$S_{\hat{x}_p \hat{x}_p}(f) = |-j\mathrm{sgn}(f)|^2 S_{x_p x_p}(f) = S_{x_p x_p}(f) \Rightarrow R_{\hat{x}_p \hat{x}_p}(\tau) = R_{x_p x_p}(\tau) \quad (2)$$

$$S_{\hat{x}_p x_p}(f) = -j\mathrm{sgn}(f)\, S_{x_p x_p}(f) \Rightarrow R_{\hat{x}_p x_p}(\tau) = \widehat{R}_{x_p x_p}(\tau) \quad (3)$$

$$S_{x_p \hat{x}_p}(f) = j\mathrm{sgn}(f)\, S_{x_p x_p}(f) \Rightarrow R_{x_p \hat{x}_p}(\tau) = -\widehat{R}_{x_p x_p}(\tau) \quad (4)$$

  with $\widehat{R}_{x_p x_p}(\tau)$ denoting the Hilbert transform of $R_{x_p x_p}(\tau)$.
  At this point, we can calculate the autocorrelation function (time-lag) of $x_c(t)$.

Teorema: se $X_p(t)$ è WSS a media nulla,

allora le sue componenti in banda base $X_c(t)$ e $X_s(t)$ sono congiuntamente WSS a media nulla

Dim

Verifico che $X_c(t)$ e $X_s(t)$ siano a media nulla

- per ipotesi $E[X_p(t)] = \mu_{X_p} = 0 \quad \forall t \in \mathbb{R}$
- $\mu_{\hat{X}_p} = E\left[\hat{X}(t) * \frac{1}{\pi t}\right] = E\left[\int_{-\infty}^{+\infty} X_p(t-\tau)\cdot\frac{1}{\pi\tau}\,d\tau\right] = \int_{-\infty}^{+\infty} E[X_p(t-\tau)]\,\frac{1}{\pi\tau}\,d\tau = 0$

$E[X_p(t)] = 0$ per ipotesi

quindi sia X_c sia X_s sono a media nulla.

→ La media è indipendente da t

Rimane da verificare che la funzione di autocorrelazione $R_{X_cX_c}(t,\tau)$ e $R_{X_sX_s}(t,\tau)$ e la funzione di mutua correlazione $R_{X_cX_s}(t,\tau)$ dipendano unicamente dal ritardo τ

Prima deriviamo delle relazioni fondamentali tra il processo reale $X_p(t)$ e la sua trasformata di Hilbert

Ricordando che $S_{yy}(f) = S_{xx}(f)\,|H(f)|^2$ e che $R_{yy}(\tau) = R_{xx}(\tau) * h_z(\tau) * h_z(-\tau)$

$$S_{X_pX_p}(f) = \underbrace{|-j\,\mathrm{sgn}(f)|^2}_{=1} S_{X_pX_p}(f) = S_{X_pX_p}(f) \xrightarrow{\mathcal{F}^{-1}} R_{\hat{X}_p\hat{X}_p}(\tau) = R_{X_pX_p}(\tau)$$

$$S_{\hat{X}_pX_p}(f) = -j\,\mathrm{sgn}(f)\,S_{X_pX_p}(f) \xrightarrow{\mathcal{F}^{-1}} R_{\hat{X}_pX_p}(\tau) = \hat{R}_{XX}(\tau)$$

corrisponde alla trasformata di Hilbert della funzione di autocorrelazione di $X_p(t)$

$$S_{X_p\hat{X}_p}(\tau) = j\,\mathrm{sgn}(f)\,S_{X_pX_p}(f) \xrightarrow{\mathcal{F}^{-1}} R_{X_p\hat{X}_p}(\tau) = -\hat{R}_{XX_p}(\tau)$$

dove $\hat{R}_{X_pX_p}(\tau)$ corrisponde alla trasformata di Hilbert di $R_{XX_p}(\tau)$

A questo punto possiamo calcolare la funzione di autocorrelazione di $X_c(t)$

$$2R_{X_cX_c}(t,\tau) = 2E[X_c(t)X_c(t-\tau)] = E\left[\left(x_p\cos(2\pi f_c t) + \hat{x}(t)\sin(2\pi f_c t)\right)\left(x_p(t-\tau)\cos(2\pi f_c(t-\tau)) + \hat{x}_p(t-\tau)\sin(2\pi f_c(t-\tau))\right)\right] =$$

$$X_c(t) = \frac{1}{\sqrt{2}}\left(X_p(t)\cos(2\pi f_c t) + \hat{X}_p(t)\sin(2\pi f_c t)\right)$$

$$= E\Big[X_p(t)X_p(t-\tau)\cos(2\pi f_c t)\cos(2\pi f_c(t-\tau)) + X_p(t)\hat{X}_p(t-\tau)\cos(2\pi f_c t)\sin(2\pi f_c(t-\tau)) + \hat{X}_p(t)X_p(t-\tau)\sin(2\pi f_c t)\cos(2\pi f_c(t-\tau)) +$$

$$+ \hat{X}_p(t)\hat{X}(t-\tau)\sin(2\pi f_c t)\sin(2\pi f_c(t-\tau))\Big] =$$

$$= R_{X_pX_p}(\tau)\cos(2\pi f_c t)\cos(2\pi f_c(t-\tau)) + R_{X_p\hat{X}_p}(\tau)\cos(2\pi f_c t)\sin(2\pi f_c(t-\tau)) + \hat{R}_{\hat{X}_pX_p}(t)\sin(2\pi f_c t)\cos(2\pi f_c(t-\tau)) +$$

$$+ R_{\hat{X}_pX_p}(\tau)\sin(2\pi f_c t)\sin(2\pi f_c(t-\tau))$$

$R_{\hat{X}_pX_p}(\tau) \to R_{X_pX_p}(\tau)$

$R_{X_p\hat{X}_p}(\tau) \to -\hat{R}_{X_pX_p}(\tau)$

$= R_{X_pX_p}(\tau)$

$$\Rightarrow 2R_{X_cX_c}(t,\tau) = R_{X_pX_p}(\tau)\left[\cos(2\pi f_c t)\cos(2\pi f_c(t-\tau)) + \sin(2\pi f_c t)\sin(2\pi f_c(t-\tau))\right] + \hat{R}_{X_pX_p}(\tau)\left[\sin(2\pi f_c t)\cos(2\pi f_c(t-\tau)) - \cos(2\pi f_c t)\sin(2\pi f_c(t-\tau))\right] =$$

$$= R_{X_pX_p}(\tau)\cos(2\pi f_c\tau) + \tilde{R}_{X_pX_p}(\tau)\sin(2\pi f_c\tau) = 2R_{X_cX_c}(\tau) \leftarrow \text{DIPENDE UNICAMENTE DAL RITARDO}$$

utilizzando le formule di Werner

$$\sin(\alpha)\cos(\beta) = \frac{1}{2}\left[\sin(\alpha+\beta) + \sin(\alpha-\beta)\right]$$

$$\cos(\alpha)\cos(\beta) = \frac{1}{2}\left[\cos(\alpha+\beta) + \cos(\alpha-\beta)\right]$$

$$\sin(\alpha)\sin(\beta) = \frac{1}{2}\left[\cos(\alpha+\beta) - \cos(\alpha-\beta)\right]$$

Ripetendo gli stessi passaggi per $X_s(t)$ e per la mutua correlazione $R_{X_cX_s}$ si ricava:

$$2R_{X_sX_s}(t,\tau) = R_{X_pX_p}(\tau)\cos(2\pi f_c\tau) + \hat{R}_{X_pX_p}(\tau)\sin(2\pi f_c\tau) = R_{X_sX_c}(\tau) = 2R_{X_cX_c}(\tau)$$

$$2R_{X_cX_s}(t,\tau) = R_{X_pX_p}(\tau)\sin(2\pi f_c\tau) + \hat{R}_{X_pX_p}(\tau)\cos(2\pi f_c\tau) = 2R_{X_cX_s}(\tau)$$

$$2R_{X_sX_c}(t,\tau) = -R_{X_pX_p}(\tau)\sin(2\pi f_c\tau) + \hat{R}_{X_pX_p}(\tau)\cos(2\pi f_c\tau) = 2R_{X_sX_c}(\tau) = -2R_{X_cX_s}(\tau)$$

Quindi per un qualsiasi processo $X(t)$ in BANDA PASSANTE che sia WSS

$$R_{X_cX_c}(\tau) = R_{X_sX_s}(\tau) \quad e \quad R_{X_cX_s}(\tau) = -R_{X_sX_c}(\tau)$$

Riprendendo ora definizione dell'inviluppo complesso: $X(t) = X_c(t) + jX_s(t)$, calcolo la ACF

$$R_{XX}(\tau) \triangleq E\left[X(t)\,X^*(t-\tau)\right] = E\left[X_c(t)X_c(t-\tau) - jX_c(t)X_s(t-\tau) + jX_s(t)X_c(t-\tau) + X_s(t)X(t-\tau)\right] =$$

(X^*: coniugato poiché X è complesso)

$$= R_{X_cX_c}(\tau) + \underbrace{R_{X_sX_s}(\tau)}_{=R_{X_cX_c}(\tau)} + j\left[\underbrace{-R_{X_cX_s}(\tau)}_{=-R_{X_sX_c}(\tau)} + R_{X_sX_c}(\tau)\right] = \underbrace{2R_{X_cX_c}(\tau)}_{\text{autocorrelazione di } X_c} + j2\underbrace{R_{X_sX_c}(\tau)}_{\text{mutua correlazione } X_s, X_c}$$

$$\Rightarrow R_{XX}(\tau) = 2R_{X_cX_c}(\tau) + j2R_{X_sX_c}(\tau)$$

Poiché $R_{XX}(0) = E\left[|X(t)|^2\right] = P_{X_p}$ è una quantità reale $\Rightarrow R_{X_cX_s}(0) = 0 = R_{X_sX_c}(0)$

$$C_{X_cX_s}(0) = R_{X_cX_s}(0) - \underbrace{\mu_{X_s}\mu_{X_c}}_{=0} = 0$$

Significa che per qualsiasi istante temporale t_0, le variabili aleatorie a media nulla $X_c(t_0)$ e $X_s(t_0)$ sono INCORRELATE per $\tau = 0$

$\Rightarrow$ $X_c(t_0, 0)$ e $X_s(t_0, 0)$ sono molteplici statisticamente indipendenti per ogni t_0 se sono Gaussiane

Representation of passband stochastic processes (3/4)

$$2\,R_{x_c x_c}(t,\tau) = 2\,\mathrm{E}[x_c(t)\,x_c(t-\tau)] = R_{x_p x_p}(\tau)\,\cos(2\pi f_c t)\,\cos[2\pi f_c(t-\tau)] + R_{\hat{x}_p \hat{x}_p}(\tau)\,\sin(2\pi f_c t)\,\sin[2\pi f_c(t-\tau)] + R_{x_p \hat{x}_p}(\tau)\,\cos(2\pi f_c t)\,\sin[2\pi f_c(t-\tau)] + R_{\hat{x}_p x_p}(\tau)\,\sin(2\pi f_c t)\,\cos[2\pi f_c(t-\tau)] \quad (5)$$

Substituting into (5) equations (2), (3) and (4), one obtains

$$2\,R_{x_c x_c}(t,\tau) = R_{x_p x_p}(\tau)\,\{\cos(2\pi f_c t)\,\cos[2\pi f_c(t-\tau)] + \sin(2\pi f_c t)\,\sin[2\pi f_c(t-\tau)]\} + \widehat{R}_{x_p x_p}(\tau)\,\{\sin(2\pi f_c t)\,\cos[2\pi f_c(t-\tau)] - \cos(2\pi f_c t)\,\sin[2\pi f_c(t-\tau)]\} = R_{x_p x_p}(\tau)\,\cos(2\pi f_c \tau) + \widehat{R}_{x_p x_p}(\tau)\,\sin(2\pi f_c \tau) \equiv 2\,R_{x_c x_c}(\tau) \quad (6)$$

To complete the proof, we can observe that

$$2\,R_{x_s x_s}(t,\tau) = R_{x_p x_p}(\tau)\,\cos(2\pi f_c \tau) + \widehat{R}_{x_p x_p}(\tau)\,\sin(2\pi f_c \tau) = 2\,R_{x_c x_c}(\tau) \quad (7)$$

$$2\,R_{x_c x_s}(t,\tau) = R_{x_p x_p}(\tau)\,\sin(2\pi f_c \tau) - \widehat{R}_{x_p x_p}(\tau)\,\cos(2\pi f_c \tau) \equiv 2\,R_{x_c x_s}(\tau) \quad (8)$$

$$2\,R_{x_s x_c}(t,\tau) = -R_{x_p x_p}(\tau)\,\sin(2\pi f_c \tau) + \widehat{R}_{x_p x_p}(\tau)\,\cos(2\pi f_c \tau) = -2\,R_{x_c x_s}(\tau) \quad (9)$$

■

Representation of passband stochastic processes (4/4)

- We thus have that a WSS passband process is such that

$$R_{x_c x_c}(\tau) = R_{x_s x_s}(\tau) \qquad R_{x_c x_s}(\tau) = -R_{x_s x_c}(\tau) \tag{10}$$

- Relying on (1), the acf of the complex envelope $x(t)$ associated to the passband process $x_p(t)$ is given by

$$R_{xx}(\tau) \stackrel{\triangle}{=} \mathrm{E}[x(t)x^*(t-\tau)] = R_{x_c x_c}(\tau) + R_{x_s x_s}(\tau) + j[R_{x_s x_c}(\tau) - R_{x_c x_s}(\tau)] \tag{11}$$

$$= 2\,R_{x_c x_c}(\tau) + 2j\,R_{x_s x_c}(\tau) \tag{12}$$

- Since $R_{xx}(0) = \mathrm{E}[|x(t)|^2]$ is a real quantity, it results $R_{x_c x_s}(0) = 0$: this means that at any time instant t_0, the zero-mean random variables $x_c(t_0)$ and $x_s(t_0)$ are *uncorrelated* $\Rightarrow$ $x_c(t_0)$ and $x_s(t_0)$ are statistically independent for any t_0 if they are Gaussian.

¿Perché?

Power spectral density of the quadrature components

- Invoking the Wiener-Khinchin theorem, we can compute the PSD of the quadrature components by taking the Fourier transform of (6) and (7).
- For the in-phase component it results:

$$
\begin{aligned}
S_{x_c x_c}(f) = \frac{1}{2} \Bigg[& \frac{S_{x_p x_p}(f - f_c) + S_{x_p x_p}(f + f_c)}{2} \\
& + \frac{-j\, S_{x_p x_p}(f - f_c)\,\mathrm{sgn}(f - f_c) + j\, S_{x_p x_p}(f + f_c)\,\mathrm{sgn}(f + f_c)}{2j} \Bigg] \\
= \frac{1}{2} & \left[S_{x_p x_p}(f - f_c)\,\frac{1 - \mathrm{sgn}(f - f_c)}{2} + S_{x_p x_p}(f + f_c)\,\frac{1 + \mathrm{sgn}(f + f_c)}{2} \right] \\
= \frac{1}{2} & \left[S_{x_p x_p}(f - f_c)\,\mathrm{u}(-f + f_c) + S_{x_p x_p}(f + f_c)\,\mathrm{u}(f + f_c) \right] \qquad (13)
\end{aligned}
$$

- The cross-spectral densities of the two random processes $x_c(t)$ and $x_s(t)$ can be obtained by performing the Fourier transform of (8) and (9).

Possiamo ol dominio deeca frequenza sfurrando c'espressione di $R_{x_c x_c}(\tau)$ e ic Teozema di Weiner-Khinchim

Conoscendo: $R_{x_c x_c}(\tau) = \frac{1}{2}\left[R_{x_p x_p}(\tau) \cos(2\pi f_c \tau) + \hat{R}_{x_p x_p}(\tau) \sin(2\pi f_c \tau) \right]$

Voeariamo ea PSD $S_{x_c x_c}(f) = \mathcal{F}\{ R_{x_c x_c}(\tau) \}$

$$S_{X_c X_c}(f) = \frac{1}{2}\left[\frac{S_{x_p x_p}(f-f_c) + S_{x_p x_p}(f+f_c)}{2} + \frac{-\jmath \operatorname{sgm}(f-f_c) S_{x_p x_p}(f-f_c) + \jmath \operatorname{sgm}(f+f_c) \cdot S_{x_p x_p}(f+f_c)}{2\jmath} \right] =$$

$$= \frac{1}{2}\left[S_{x_p x_p}(f-f_c)\left(\frac{1 - \operatorname{sgm}(f-f_c)}{2} \right) + S_{x_p x_p}(f+f_c)\left(\frac{1 + \operatorname{sgm}(f+f_c)}{2} \right) \right] =$$

$$1 - \operatorname{sgm}(f-f_c) = \begin{cases} 0 & f > f_c \\ -2 & f < f_c \end{cases} = 2u(-f+f_c)$$

$$1 + \operatorname{sgm}(f+f_c) = \begin{cases} 2 & f > -f_c \\ 0 & f < -f_c \end{cases} = 2u(f+f_c)$$

$$= \frac{1}{2} S_{x_p x_p}(f-f_c)\, u(-f+f_c) + \frac{1}{2} S_{x_p x_p}(f+f_c)\, u(f+f_c)$$

Dato che $R_{x_c x_c}(\tau) = R_{x_s x_s}(\tau) \Rightarrow S_{x_c x_c}(f) = S_{x_s x_s}(f)$

La mutua densità spettrale di potenza (cross-PSD) tra $x_c(t)$ e $x_s(t)$ ea possiamo ottenere sfurrando:

$$R_{x_c x_s}(\tau) = -R_{x_s x_c}(\tau)$$

$$2R_{x_c x_s}(t,\tau) = R_{x_p x_p}(\tau)\sin(2\pi f_c \tau) - \hat{R}_{x_p x_p}(\tau)\cos(2\pi f_c \tau) \equiv 2R_{x_c x_s}(\tau)$$

$$2R_{x_s x_c}(t,\tau) = -R_{x_p x_p}(\tau)\sin(2\pi f_c \tau) + \hat{R}_{x_p x_p}(\tau)\cos(2\pi f_c \tau) = -2R_{x_c x_s}(\tau)$$

An important remark (1/2)

- The analytical signal is obtained by filtering $x_p(t)$ through $H(f) = \sqrt{2}\,\mathrm{u}(f)$ (see eq. (3) of the lecture on Rice representation for deterministic signals), hence its PSD is

$$S_{x_p^+ x_p^+}(f) = S_{x_p x_p}(f)\,|H(f)|^2 = 2\,S_{x_p x_p}(f)\,\mathrm{u}(f)$$

- Since $x(t) = x_p^+(t)\,e^{-j2\pi f_c t}$, one has

$$R_{xx}(\tau) = \mathrm{E}[x(t)\,x^*(t-\tau)] = R_{x_p^+ x_p^+}(\tau)\,e^{-j2\pi f_c \tau}$$

  i.e., multiplication by a complex exponential preserves wide-sense stationarity.

- As a consequence, the PSD of the complex envelope is the version of $S_{x_p^+ x_p^+}(f)$ translated around the origin:

$$S_{xx}(f) = S_{x_p^+ x_p^+}(f + f_c) = 2\,S_{x_p x_p}(f + f_c)\,\mathrm{u}(f + f_c) \tag{14}$$

Il processo aleatorio ANALITICO $X_p^+(t)$ si ottiene attraverso il passaggio in un filtro con $H(f) = \sqrt{2}\,u(f)$

Ricordando la relazione input-output per la PSD in un filtro LTI

$$S_{out}(f) = S_{in}(f)\,|H(f)|^2$$

Quindi:

$$S_{X_p^+ X_p^+}(f) = S_{X_p X_p}(f) \cdot |H(f)|^2 = S_{X_p X_p}(f)\, 2\,u(f)$$

il segnale analitico annulla completamente le frequenze negative, e raddoppia la potenza delle frequenze positive

- Partendo dalla definizione dell'inviluppo complesso $X(t) = X_p^+(t)\, e^{-j2\pi f_c t}$,
si ha che:

$$R_{XX}(\tau) = E[X(t)X^*(t-\tau)] = E\left[X_p^+(t)\, e^{-j2\pi f_c t} X_p^{+*}(t-\tau)\, e^{+j2\pi f_c (t-\tau)}\right] = E\left[X_p^+(t) X_p^+(t-\tau)\right] e^{-j2\pi f_c t + j2\pi f_c (t-\tau)} =$$

$$= R_{X_p^+ X_p^+}(\tau)\, e^{-j2\pi f_c \tau}$$ ← dipende unicamente dal ritardo

$\Rightarrow$ la moltiplicazione per un esponente complesso preserva la stazionarietà in senso lato (WSS)

Come conseguenza la densità spettrale di potenza (PSD) dell'inviluppo complesso:

$$S_{XX}(f) = \mathcal{F}\{R_{XX}(\tau)\} = \mathcal{F}\{R_{X_p^+ X_p^+}(\tau)\, e^{-j2\pi f_c \tau}\} = S_{X_p^+ X_p^+}(f + f_c)$$ ← PSD traslata intorno all'origine

Corrisponde alla versione traslata della PSD del segnale analitico verso sinistra di f_c

In conclusione, per ottenere la PSD dell'inviluppo complesso del processo passa banda bisogna

- Valutare la PSD del segnale passa banda $S_{X_p X_p}(f)$
- Eliminare attraverso un filtro le frequenze negative $H(f) = \sqrt{2}\, u(f)$ e moltiplicarle per $\sqrt{2}$ (permette di preservare la potenza originaria del segnale)

$$S_{X_p^+ X_p^+}(f) = S_{X_p X_p}(f)\,|H(f)|^2 = 2 S_{X_p X_p}(f)\, u(f)$$

- Traslare a sinistra di f_c portando la PSD del segnale analitica centrata nell'origine

$$S_{XX}(f) = S_{X_p^+ X_p^+}(f + f_c) = 2 S_{X_p X_p}(f + f_c)\, u(f + f_c)$$

An important remark (2/2)

- If the PSD of a passband random process is *symmetric* around f_c, from (14) the PSD of the complex envelope $x(t)$ is an *even* function $\Rightarrow$ $R_{xx}(\tau)$ is real for all τ $\Rightarrow$ from (12) it results that

$$R_{x_s x_c}(\tau) = 0 \quad \text{for all } \tau$$

- Consequently, $x_c(t)$ and $x_s(t)$ are incoherent (and independent when $x_p(t)$ is Gaussian).

- In this situation, $S_{xx}(f) = 2\,S_{x_c x_c}(f) = 2\,S_{x_s x_s}(f)$.

Caso particolare molto frequente nei sistemi di trasmissione:

> La PSD del processo reale $S_{X_pX_p}(f)$ è simmetrica rispetto la frequenza portante f_c

Dalla relazione $S_{xx}(f) = 2\, S_{x_px_p}(f+f_c)\, u(f+f_c)$

Posso dedurre che $\boxed{S_{xx}(-f) = S_{xx}(f)}$ ovvero che la PSD dell'inviluppo complesso è PARI

$$S_{xx}(-f) = 2S_{x_px_p}(-f+f_c)\, u(-f+f_c) = 2\, S_{X_pX_p}(f+f_c)\, u(-f+f_c) =$$

($S_{X_pX_p}$ è PARI rispetto f_c; $u(-f+f_c)$)

Essendo $S_{xx}(f) = \mathcal{F}\{R_{xx}(\tau)\}$, per la proprietà della trasformata di Fourier

l'antitrasformata di una funzione pari e reale è anche una pari e reale

$\Rightarrow R_{xx}(\tau)$ è REALE e PARI

$\Rightarrow \mathrm{Im}\{R_{xx}(\tau)\} = 0 \quad \forall \tau \in \mathbb{R}$

Dato che:

$$R_{xx}(\tau) = 2R_{x_cx_c}(\tau) + j\,2R_{x_sx_c}(\tau)$$

(mutua correlazione x_s, x_c)

$$\Rightarrow R_{x_sx_c}(\tau) = 0 \Rightarrow -R_{x_cx_s}(\tau) = 0$$

(Considerando $X_p(t)$ WSS)

Ciò quindi significa che $x_c(t)$ e $x_s(t)$ sono INCORRELATI

-> Se consideriamo anche $X_p(t)$ a media nulla $\Rightarrow$ INCORRELATE $\Rightarrow$ STATISTICAMENTE INDIPENDENTI

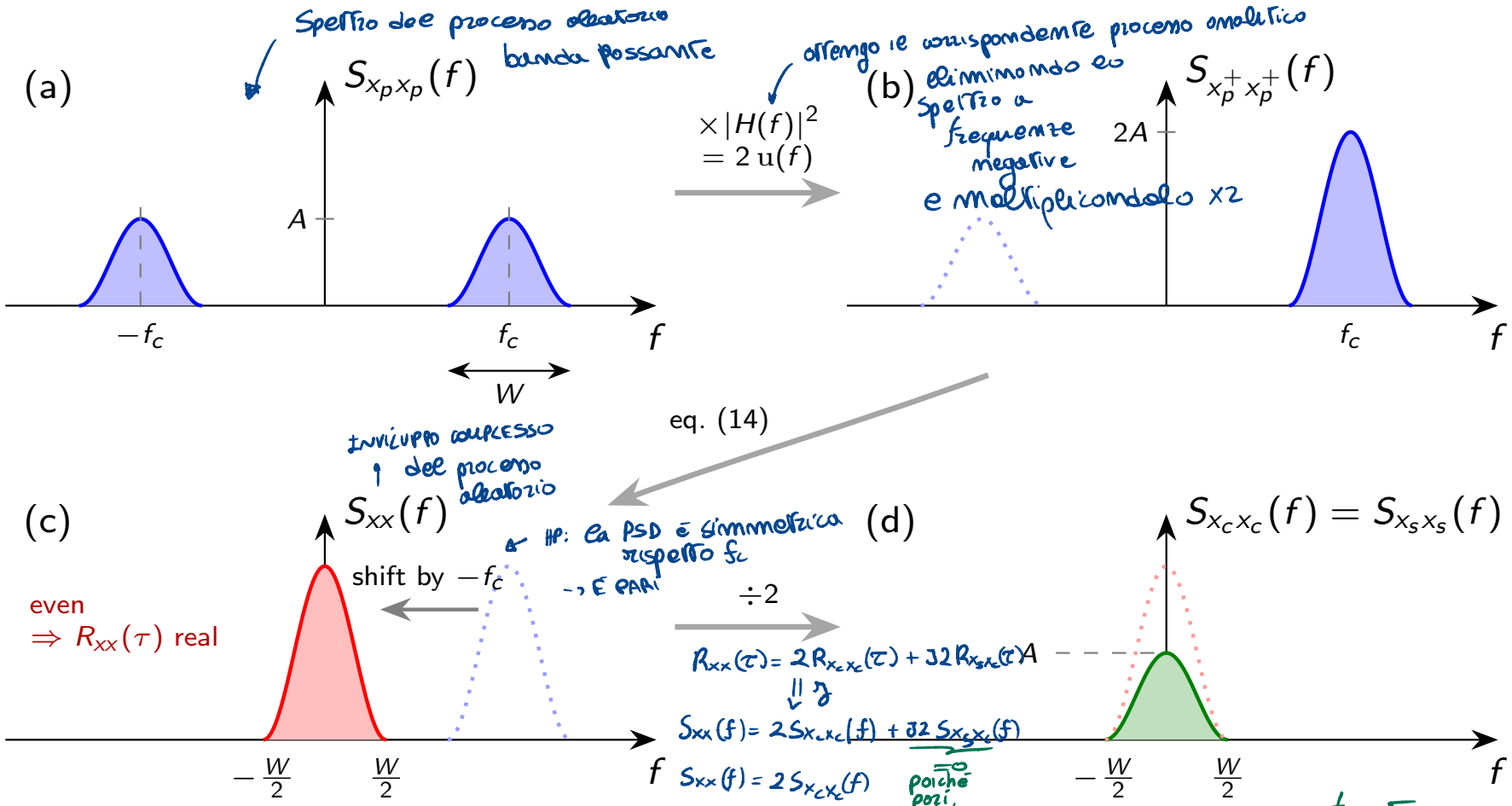
(supponendo Gaussiane)

Mentre l'incoerenza, in generale, è garantita solo nel singolo istante ($\tau = 0$),

quando lo spettro è simmetrico attorno a f_c le due componenti $x_c(t)$ e $x_s(t)$

sono completamente incoerenti/incorrelate $\forall \tau \in \mathbb{R}$

An important remark: a graphical view



If $S_{x_p x_p}(f)$ is symmetric around f_c, then $S_{xx}(f)$ is even $\Rightarrow R_{x_s x_c}(\tau) = 0$ and $S_{xx}(f) = 2\,S_{x_c x_c}(f) = 2\,S_{x_s x_s}(f)$.

An example: in-band noise

- Let us consider a passband noise $w_p(t)$ having constant PSD $S_{w_p w_p}(f) = N_0/2$ over a given bandwidth W.
- We now calculate the PSDs $S_{w_c w_c}(f)$ and $S_{w_s w_s}(f)$ of the two quadrature components $w_c(t)$ and $w_s(t)$. It is easily seen that

$$S_{w_c w_c}(f) = S_{w_s w_s}(f) = \frac{N_0}{2} \quad \text{for } |f| < \frac{W}{2}$$

- Moreover, the PSD of the complex envelope is $S_{ww}(f) = N_0$ for $|f| < W/2$.
- It is worth noting that the power of the original passband process is $P_{w_p} = 2W(N_0/2) = N_0\, W$ and coincides with the power of the complex envelope.
- Conditions (10) are of particular importance: the correlation properties of the complex envelope of any passband WSS process can be specified by the acf of the in-phase component and the cross-correlation between the quadrature components.
- Additionally, the in-phase and the quadrature components in the same instant are uncorrelated $\Rightarrow$ the quadrature components of a complex Gaussian process in the same time instant are statistically independent.

IL RUMORE IN BANDA PASSANTE

$R_{w_p w_p}(\tau) = \frac{N_0}{2}\delta(\tau)$ → Γ

Consideriamo un rumore bonda passante $w_p(t)$ avente una PSD costante $S_{w_p w_p}(f) = \frac{N_0}{2}$ con bonda W

Il rumore bianco gaussiano (AWGN) potrebbe essere un rumore del genere, poiché ha PSD costante $S(f) = \frac{N_0}{2}$ su tutte le frequenze $(-\infty, +\infty)$

=> potenza media infinita

Nei sistemi reali il ricevitore filtra il segnale in ingresso con un filtro di bonda W centrato su f_c

=> ottenendo così un rumore con:

- $S(f) = \frac{N_0}{2}$
- bonda pari a W

w_c e w_s sono componenti in quadratura

CALCOLO DELLE PSD IN BANDA BASE del rumore: $S_{w_c w_c}(f)$ e $S_{w_s w_s}(f)$

=> Essendo che $S_{ww}(f) = 2 S_{w_p w_p}(f-f_c)\,u(f-f_c) = 2\frac{N_0}{2}u(f-f_c) = N_0\,u(f-f_c)$ per $|f| < \frac{W}{2}$

Essendo $S_{x_p x_p}(f)$ reale e costante => è pari rispetto a bonda

=> $S_{ww}(f) = 2 S_{w_c w_c}(f)$

~> $S_{w_c w_c}(f) = S_{w_s w_s}(f) = \frac{N_0}{2}$ per $|f| < \frac{W}{2}$

→ È utile notare che la potenza del segnale originale

$$P_{w_p} = E\left[|X_p(t)|^2\right] = \frac{N_0}{2} \cdot 2W = N_0 W$$

coincide con la potenza dell'inviluppo complesso

Considerando che, le componenti in fase w_c e in quadratura x_s nello stesso istante t sono incorrelate

$\Rightarrow$ Queste sono statisticamente indipendenti se si tratta di un processo Gaussiano

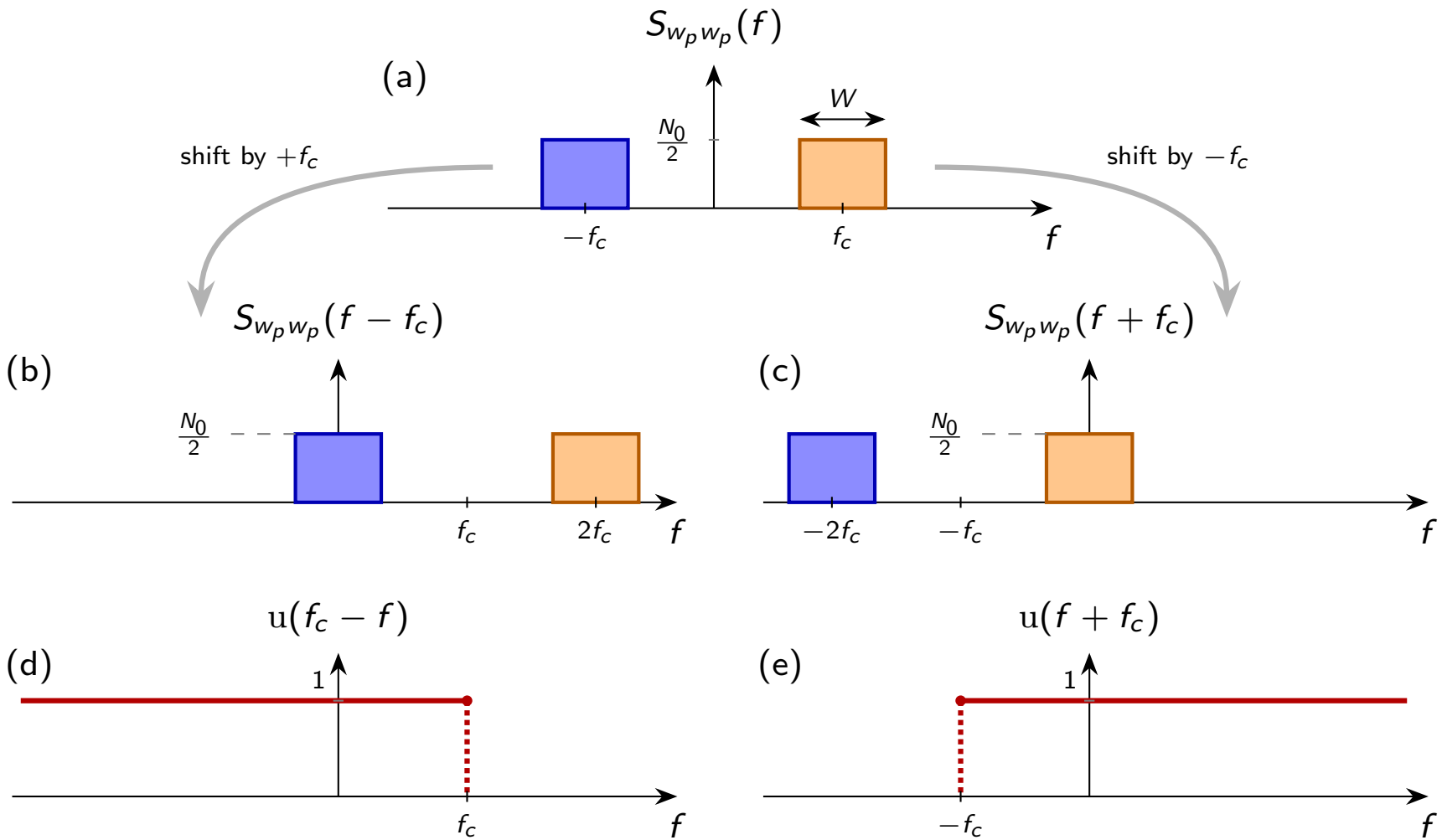
$\Rightarrow$ essendo $S_{ww}(f)$ è reale $\Rightarrow$ $S_{w_c w_s}(f) = 0 \quad \forall f \in \mathbb{R}$

$\Rightarrow$ Sono INCORRELATE $\Rightarrow$ STATISTICAMENTE INDIPENDENTI

↑ se w_p è Gaussiano

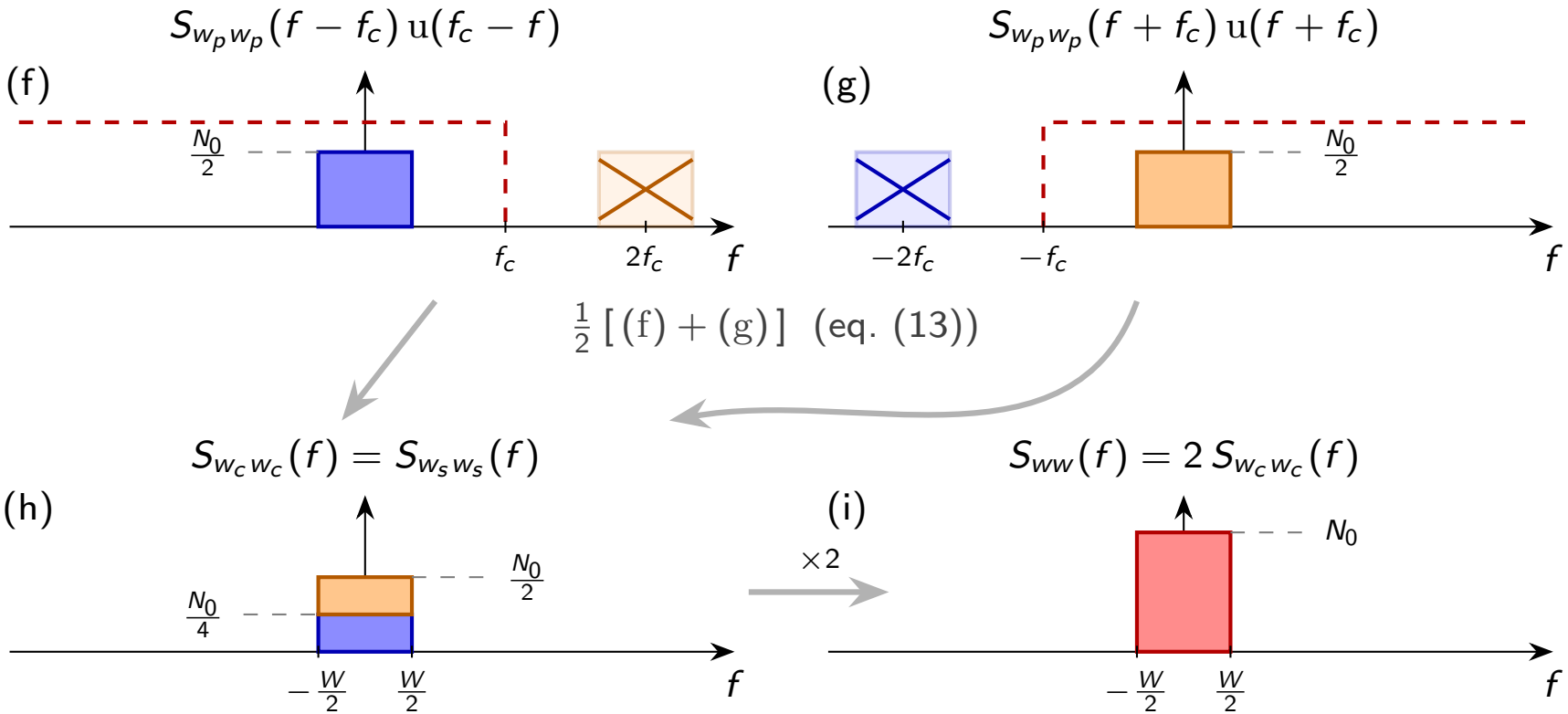
An example: in-band noise – graphical construction of $S_{w_c w_c}(f)$ (1/2)

- Applying (13) to $S_{w_p w_p}(f)$: first shift the passband PSD by $\pm f_c$, then multiply by the unit steps $\mathrm{u}(f_c - f)$ and $\mathrm{u}(f + f_c)$.



An example: in-band noise – graphical construction of $S_{w_c w_c}(f)$ (2/2)

- Each unit step retains only the lobe falling around $f = 0$; summing and halving gives $S_{w_c w_c}(f) = S_{w_s w_s}(f) = N_0/2$ for $|f| < W/2$.



Since $S_{w_p w_p}(f)$ is symmetric around f_c: $S_{ww}(f) = 2\,S_{w_c w_c}(f) = N_0$ for $|f| < W/2$, and $P_{w_p} = N_0 W = P_w$.